

KAVYA PARIVARA BABU

Cary, NC | kavyababu0625@gmail.com | +1-919-200-8248
[linkedin.com/in/kavya-pb](https://www.linkedin.com/in/kavya-pb)

EDUCATION

Master of Science: Information Systems Northwest Missouri State University, Maryville, Missouri	Aug 2023 - Dec 2024 GPA 4/4
Bachelor of Engineering: Electronics and Communication Canara Engineering College, Mangalore, India	Jun 2008 - May 2012 GPA 3.25/4

SUMMARY

- Skilled Java Developer with over 8 years of professional experience in designing, developing, implementing, and testing client-server web applications.
- Expertise in Java/J2EE, Spring Framework, Spring MVC, Spring Boot, Spring Security, Spring Data JPA, Hibernate, Microservices, Webservices, RESTful APIs, Kafka Messaging, Java Messaging Service (JMS), GCP Pub/sub messaging, and JDBC Template.
- Proficient in Google Cloud Platform (GCP) with hands-on experience in GCP services, including expertise in Pub/Sub messaging for real-time data streaming and event-driven architectures.
- Familiar with Python basics, including scripting, data manipulation, and automation.
- Proficient in working with relational databases like Oracle SQL, MySQL, and Impala, and familiar with NoSQL databases such as MongoDB.
- Designed and implemented secure authentication and authorization mechanisms using OAuth 2.0 and JWT to protect APIs and enhance security across web applications.
- Proficient in developing dynamic and responsive web applications using HTML, CSS, and JavaScript, integrated with Spring Boot MVC architecture to ensure scalable, secure, and maintainable full-stack solutions.
- Experienced in using Apache Maven for managing build processes, dependency management, and automating project lifecycles in Java-based applications.
- Expertise in writing JUnit test cases to ensure robust and reliable code coverage for various software components.
- Expertise in Jira, Confluence, Bitbucket, TeamCity, GitHub, Git clients, and Veracode Greenlight, with hands-on experience integrating Jira for project tracking and issue management, and Confluence for documentation and collaboration to streamline workflows and enhance team productivity.
- Proficient in designing and implementing CI/CD workflows using tools like GitHub Actions, Azure DevOps, Jenkins, and TeamCity to automate build, test, and deployment processes, ensuring faster and more reliable software delivery. Experienced with containerization and orchestration technologies such as Docker, Docker Hub, Kubernetes, and OpenShift.
- Skilled in optimizing CI/CD pipelines, including the use of JFrog Artifactory for secure artifact storage and version control, and deploying microservices to OpenShift clusters via OpenShift Console, improving deployment scalability and reliability
- Proficient in utilizing Splunk for real-time monitoring, log analysis, and alerting to ensure system reliability and performance.
- Strong understanding of caching strategies and memory management in Hazelcast.
- Proficient in the complete Software Development Life Cycle (SDLC), from requirements gathering to delivery, using both Waterfall and Agile methodologies.
- Experienced in the Insurance, Banking, and E-commerce domains.

WORK EXPERIENCE

Graduate Assistant Northwest Missouri State University Maryville, MO	Aug 2024 - Dec 2024
• Assisted the course instructor in delivering and managing a curriculum focused on Information Systems Analysis and Design, emphasizing agile methodologies.	

- Facilitated in-class exercises on use case generation and data flow diagrams (DFDs), providing practical guidance.
- Utilized MongoDB to design and manage NoSQL databases for academic projects while supporting faculty and students as a Graduate Assistant.
- Organized help sessions to resolve queries on assignments, group projects, and theoretical concepts, promoting clarity and application.
- Guided students in creating industry-standard models, including use case diagrams, class diagrams, and UI designs.
- Evaluated assignments, exams, and projects, providing feedback to measure and improve student outcomes.
- Collaborated in preparing course materials such as lecture notes and rubrics, ensuring clear and consistent delivery.

Senior System Analyst

Sep 2021 - Jul 2023

IBM | Bangalore, India

- Designed, implemented, supported, and maintained Java Spring Boot microservice applications, adhering to best practices for reliability, scalability, and security.
- Excelled in migrating existing messaging infrastructure from Kafka to Google Cloud Pub/Sub using GCP Pub/Sub channels to eliminate the boilerplate code, simplify scaling, and seamless GCP integration, leading to a notable 20% to 30% reduction in operating costs.
- Utilized JWT (JSON Web Token) to manage stateless authentication, ensuring secure API communications by encrypting user session data.
- Utilized JMeter to conduct comprehensive performance testing of application endpoints.
- Delivered high-quality work on time by efficiently managing tasks, resolving defects, creating user support resources, and actively participating in Agile Scrum.
- Ensured code quality through rigorous testing, code reviews, and adherence to business requirements within specified timelines and budgets.
- Designed and implemented dynamic front-end features using HTML, CSS, and JavaScript, with backend services built using Spring Boot MVC, enabling seamless data flow between the UI and the server.
- Developed and maintained GitHub Actions and custom workflows to streamline CI/CD pipelines, automate code linting, unit testing, and deployment across multiple environments, resulting in a 30% reduction in deployment time and improved code quality.
- Configured and managed CI/CD pipelines using TeamCity with Bitbucket for version control to automate builds, tests, and deployments of Java Spring Boot applications. Integrated JFrog Artifactory for secure artifact management and deployed containerized applications to OpenShift clusters using OpenShift Console for monitoring and management.
- Configured New Relic to track application performance metrics, including response time, throughput, and error rates across microservices.
- Managed and triaged defects by identifying, categorizing, prioritizing, and tracking them using a defect management tracking system, ensuring effective communication between development, QA, and product teams, while maintaining detailed records of defects, root causes, and resolutions.
- Conducted Unit Testing, Integration Testing, and smoke testing post-deployment to code functionality and smooth production rollout.
- Actively participated in an Agile framework as an individual contributor, engaging in story grooming sessions, reviewing user stories, and refining acceptance criteria with the Business team.
- Received Innovation Award for creative problem-solving.

Module Lead

Apr 2019 - Sep 2021

HCL Technologies | Bangalore, India

- Wrote well-designed, efficient, and testable code using Java, Spring Boot, RESTful Web Services, Kafka and JMS Messaging Services, and Microservices, adhering to coding standards and best practices.
- Developed Kafka producers and consumers for real-time data streaming, handling high-throughput messaging scenarios across various microservices.
- Implemented JMS queues and topics for message-driven processing, ensuring secure and efficient communication within Java-based enterprise systems.

- Configured Splunk to monitor application logs in cloud-based environments, providing real-time alerts for performance issues and anomalies.
- Utilized Maven to automate build processes, manage dependencies, and optimize multi-module project structures in Java applications.
- Worked extensively with Oracle SQL for database querying, and optimization to support business-critical Java Spring Boot applications.
- Configured, developed, and optimized CI/CD pipelines using TeamCity with Bitbucket, GitHub Actions, Jenkins, and Docker to automate the build, test, and deployment processes for Java Spring Boot applications and containerized applications deployed to Kubernetes clusters.
- Resolved Java application memory leak using Eclipse Memory Analyzer tool, boosting system stability by 30%.
- Developed a Spring Boot RESTful API to submit batch Spark jobs to a YARN cluster
- Integrated Hazelcast as a distributed cache to reduce database load and improve response times for product catalog and inventory services.
- Integrated shell scripts into Control-M workflows by configuring job definitions, parameters, and variables.
- Integrated Grafana with a Spring Boot application to monitor and visualize application metrics in real-time.
- Configured Spring Boot Actuator and Micrometer to expose metrics, and set up Prometheus for metric collection.
- Developed unit tests using JUnit to ensure code quality and functionality in Java applications.
- Integrated Jira for project tracking, issue management, and sprint planning in an agile environment.
- Used Confluence to create and maintain technical documentation, including API specs, architecture diagrams, deployment processes, and troubleshooting guides.
- Mentored junior developers, conducted knowledge-sharing sessions, and provided technical guidance to foster skill development and team growth.

Senior Software Engineer

Feb 2018 - Apr 2019

Tech Mahindra | Bangalore, India

- Developed Java Spring Boot applications utilizing the Spring Framework with MVC architecture and RESTful Web services, ensuring scalability, flexibility, and maintainability.
- Configured complex business rules, workflows, and object models to implement significant enhancements to the policy administration system, and claims processing, and ensure efficient and accurate representation of Professional Liability and General Liability products.
- Optimized existing software performance, and fixed defects to enhance user experience.
- Took ownership of resolving production issues, conducting root cause analysis, and sharing findings with the team to prevent future occurrences.

Senior Systems Engineer

Jul 2013 - Feb 2018

Infosys Limited | Bangalore, India

- Developed Java/J2EE applications utilizing MVC architecture and RESTful Web services, ensuring scalability, flexibility, and maintainability.
- Performed root cause analysis, investigation, and resolution for critical problem tickets and production issues.
- Handled problem tickets, user requests, and change requests for application enhancements and bug fixes.
- Responsibly fixed production issues related to application releases.
- Wrote SQL queries for database interactions and data manipulation.
- Conducted unit testing, integration testing, and smoke testing of code changes on deployed servers.
- Consolidated and reported triage/aged/high priority defect status to onsite leads.
- Mentored and assisted other developers with implementation and debugging
- Provided production support after the product went live, ensuring smooth operations, and addressing issues promptly, and maintaining system stability and performance.
- Received Star Performer award for outstanding contributions to project delivery.

PROJECTS

Waste Management Reporting System

Sep 2024 - Dec 2024

- Designed and implemented a user-friendly UI for logging waste, adding and deleting categories, managing subcategories, and generating dynamic reports.
- Developed backend functionality with Python and Flask for saving, retrieving, editing, and deleting waste data in the SQLite database.
- Integrated authentication and authorization features using Flask sessions and bcrypt for secure user management.
- Created dynamic reporting features, including:
 - Reporting Summary Table showcasing total waste recycled, waste generated, and diversion rates.
 - Pie charts and graphical reports for waste by type and monthly trends.
 - Revenue and landfill expense tracking, with functionalities to add, delete, and export data to Excel using `xlswriter`.
- Developed a dynamic reporting system for waste management using Flask and SQLite, applying sorting algorithms to organize waste data by type and date, improving data retrieval efficiency.
- Implemented a robust date validation mechanism, ensuring admins and users can log data only for valid past or present dates.
- Conducted unit and integration testing to validate the functionality and stability of all developed features.
- Enabled Excel report generation for revenue and landfill expenses, simplifying data analysis and record-keeping.
- Successfully enabled the client to transition from a paper-based process to a digital platform for logging waste data, tracking revenue, and generating comprehensive reports, significantly enhancing operational efficiency and data management.

SKILLS

Programming Languages: Java/J2EE, Java 8, Java 17, Spring Boot, Spring Data JPA, Spring security, Spring cloud, Webservices, RESTful API's, JDBC, Microservice, API Gateway, OAuth2.0, JWT, Design patterns, SQL, Hibernate, Python, Flask, SQLAlchemy, HTML, HTML5, CSS, JSP, XML, JSON

Frameworks: Spring Framework, Spring Boot, Spring MVC, Hibernate, Mockito

Database: Oracle, MySQL, Impala, SQLite, Relational Databases, MongoDB, NoSQL

Messaging System: Apache Kafka, Messaging Queue (MQ), JMS, dbusMQ, GCP PubSub messaging

Cloud Services: Google Cloud Platform (GCP)

CI/CD Tools and Platforms: Maven, Docker, Docker Hub, Kubernetes, Jenkins, Azure DevOps, GitHub, Bitbucket, CI/CD Pipelines, GitHub Actions, GitHub Workflows, JetBrains, TeamCity, JFrog, OpenShift

Tools Used: Postman, Swagger UI, GIT, Azure DevOps, GitHub, Bit Bucket, Jira, Confluence, SonarQube, Hue –Editor, SpringToolSuite(STS), Pandas, Splunk, New Relic, Eclipse Memory Analyzer (MAT), Apache JMeter, control-M, Prometheus, Grafana, Hazelcast

Development Tools: IntelliJ, Eclipse, NetBeans, Visual Studio, MS Office Suite

Server: WebSphere Application Server (WAS), JBoss (Wildfly) Server, Apache Tomcat server